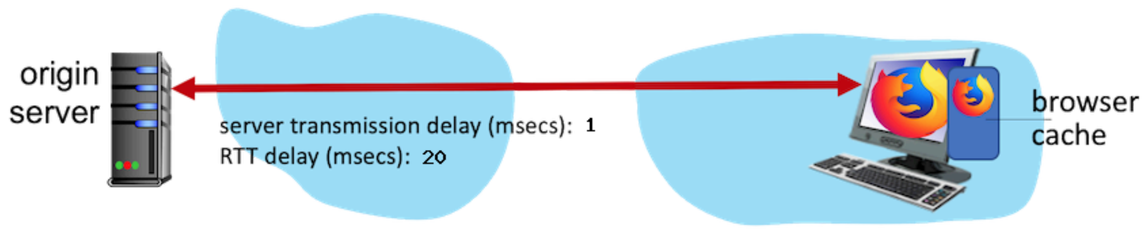
INTERACTIVE END-OF-CHAPTER EXERCISES

BROWSER CACHING

Consider an HTTP server and client as shown in the figure below. Suppose that the RTT delay between the client and server is 20 msecs; the time a server needs to transmit an object into its outgoing link is 1 msecs; and any other HTTP message not containing an object has a negligible (zero) transmission time. Suppose the client again makes 100 requests, one after the other, waiting for a reply to a request before sending the next request.



Assume the client is using HTTP 1.1 and the IF-MODIFIED-SINCE header line. Assume 50% of the objects requested have NOT changed since the client downloaded them (before these 100 downloads are performed)

QUESTION LIST

1. How much time elapses (in milliseconds) between the client transmitting the first request, and the completion of the last request?

SOLUTION

1. (RTT * NUM_PACKETS) + (NUM_PACKETS * (PERCENT__NOT_CACHED / 100) * TRANS_DELAY) = (20 * 100) + (100 * ((100-50) / 100) * 1) = 2050 ms

QUESTION 1 OF 1

How much time elapses (in milliseconds) between the client transmitting the first request, and the completion of the last request?

Answer

CHECK | HINT | SKIP

We greatly appreciate the work of John Broderick (UMass '21) in helping to develop these interactive problems.

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